USER-CENTRIC VIDEO SUMMARIZATION WITH GPT-2 AND VIT: A HYBRID APPROACH TO FRAME-LEVEL CAPTIONING

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ABSTRACT

Generic video summaries are the concise versions of lengthy videos which provide full information about the content of the whole video. However, user specific video summaries are required to produce a summary of a video customized to the user's needs. Existing models for this task do not harness the powers of pretrained huge models like blip, ViTs and LLMs and instead rely on CNNs for their proven performance in image understanding tasks. We propose a novel approach based on the combination of GPT2, ViT and LlamaIndex. A hybrid model comprising ViT and GPT2 was fine tuned in which ViT acted as image encoder for encoding image into its visual features, and GPT-2 acted as a decoder to translate the image embeddings into human understandable frame-level captions. These captions were filtered against the user input using LlamaIndex, and the corresponding frames of the resulting matched captions constitute the video summary. The experiment is carried out on videos from ego4d dataset which is the world's largest egocentric (first person) video ML dataset and benchmark suite. A subset comprising 44 videos is selected from this repository. Experimentation yielded promising results with F1 score and recall being 47.7 and 57.3 respectively through comparing the frames with ground truth summaries.

Keywords: User-specific video summaries - Pretrained models - Vision Transformers (ViTs) - Large Language Models (LLMs) - GPT-2 - LlamaIndex - Hybrid model - Frame-level captions - Ego4D dataset.

1. INTRODUCTION

The exponential growth of video content across various platforms has resulted in the demand for effective and efficient video summarization techniques. As the volume of video data grows, it has become nearly impossible for one to keep up with the latest trends, and knowledge acquisition through time-consuming videos. Traditional video summarization techniques, while effective, often fail to meet the specific needs of users, especially when summarizing content based on personal preferences or queries. This challenge has given rise to the need for user-specific video summaries, where the content is customized to the user's requirements. Therefore, we have provided a solution that revolutionizes interaction with rich and diverse video content, providing users with personalized summaries based on their written queries. The summarization process captures the most important scenes, actions, events, and content from the input video while maintaining coherence and avoiding redundancy and fulfilling all the aspects of user provided query. Existing video summarization techniques revolved around developing deep learning models from scratch and training those models on the custom dataset, but the sudden boom in Generative AI and consequent development of huge models trained on immense amounts of data suspended this approach. The reason behind this is the extreme amount of computational resources put into development of such models. Therefore, in order to leverage ourselves from pre-trained architectures' remarkable performance, we provide a one-click solution which harnesses the capabilities of Vision Transformer and GPT-2 to generate user curated video summaries.

2. RELATED WORK

The authors (SHARGHI; GONG; SHAH, 2016) introduced a novel method based on the Sequential and Hierarchical Determinantal Point Process (SH-DPP). The approach combines the relevance of video shots to user queries and their importance in the context of the entire video. By leveraging a two-layer hierarchical structure, SH-DPP ensures diversity in the selected video shots while prioritizing query relevance. Unlike previous models, SH-DPP learns from user annotations, making it adaptable to individual preferences and capable of generating personalized video summaries. Extensive experiments demonstrate that SH-DPP outperforms baseline methods on various benchmarks, including the UTE and TV episodes datasets, showing significant improvements in both hitting recall and overall summarization quality.

The paper *Query-Focused Video Summarization: Dataset, Evaluation, and A Memory Network based Approach* by (SHARGHI; LAUREL; GONG, 2017) addresses the challenge of user subjectivity in video summarization which arises from the varying preferences of users regaridng what should be included in a summary. Traditional methods focus on generic video summarization, but fail to account for the individual preferences of users, such as the specific content they are interested in within a video. To overcome this challenge, the authors proposed a memory network-based model, parameterized by a sequential determinantal point process (DPP). This model effectively attends to the user queries and select video shots that align with user's query. A significant contribution of the paper is the creation of a dense, concept-annotated video dataset that supports the evaluation of video summarization models. This method achieves an average F1 score of 44.19%, outperforming the baselines (SH-DPP(SHARGHI; GONG; SHAH, 2016) & SeqDPP(GONG et al., 2014)) by more than 10%.

The authors proposed CLIP-It, a novel framework that tackles both generic and query focused video summarization tasks in a single model. It uses a language-guided multimodal transformer to score video frames based on their importance and correlation with a user-defined query or automatically generated caption. The model can be trained with or without ground-truth supervision and achieves state-of-the-art results on standard video summarization datasets (TVSum and SumMe) and a query focused dataset (QFVS), demonstrating strong generalization capabilities. The CLIP-It method achieves an average F1 score of 54.55 on the QFVS dataset, outperforming the best baseline (QFVS(SHARGHI; LAUREL; GONG, 2017), SH-DPP(SHARGHI; GONG; SHAH, 2016) & SeqDPP(GONG et al., 2014)) by 10%, and demonstrates its ability to generate different summaries for the same video based on different input user queries. The qualitative results show that the method can effectively identify relevant frames in the video that match the input query, such as "Book and chair" or "Sun and tree", and generate summaries that are conditioned on the user's interests (NARASIMHAN; ROHRBACH; DARRELL, 2021).

Basavarajaiah M and Sharma P (ASHISH, 2017) proposed approaches which aim to model frames' dependencies using variants of Transformer Network's self attention mechanism. One of the approaches (FAJTL et al., 2019) calculates regression of the frames' importance scores by combining a soft self attention mechanism with a two-layer fully connected network. Liu et al (LIU et al., 2019) present a hierarchical based approach that first identifies the shot-level candidate key-frames, and then uses a multi-head attention model to evaluate and select key-frames for the summary. Li et al (LI et al., 2021) enhance self-attention mechanism's training pipeline by adding a step that computes attention values in order to increase the visual content diversity of the summary. These computed attention values are then used to determine the frames' importance. A variation of architecture from (FAJTL et al., 2019) is proposed by Ghauri el al (GHAURI; HAKIMOV; EWERTH, 2021), incorporating video content's additional representations. In addition to using typical CNN-based features from the pool5 layer of GoogleNet (SZEGEDY et al., 2015) trained on ImageNet, an Inflated 3D ConvNet (CARREIRA; ZISSERMAN, 2017) trained on Kinetics is also employed. These different sets of features are processed through self-attention mechanism, and the outputs from this are then combined to form a common embedding space for video frames' representation.

The authors propose UniMD, a unified framework that tackles both moment retrieval and temporal action detection tasks.The UniMD framework consists of a shared encoder that extracts features from input videos, task-specific heads for moment retrieval (MR) and temporal action detection (TAD), and multi-task learning to optimize both MR and TAD losses simultaneously. The model is trained using a multi-task loss function that combines moment retrieval and action detection losses. The UniMD model achieves competitive performance on the Charades and Charades-STA benchmarks, with a new state-of-the-art (SOTA) result of 63.98 in R1@50 for moment retrieval and significant improvement in temporal action detection when incorporating CLIP features. On the ActivityNet (ANet) and ANet-Caption benchmarks, the co-trained UniMD model sets new SOTA results, achieving 39.83 mAP and 60.29 mAP@50 in temporal action detection and 80.54 R5@50 in moment retrieval. On the Ego4D dataset the model achieves state-of-the-art results, outperforming existing methods in both Temporal Action Detection (TAD) and Moment Retrieval (MR) tasks by achieving the result of 44.8 in R1@50 for moment retrieval (ZENG et al., 2024). (MONEY; AGIUS, 2008) (BASAVARAJAIAH; SHARMA, 2021) approaches the video summarization by detecting shot boundaries or extracting keyframes (TIWARI; BHATNAGAR, 2021), reducing redundancy, and the scene changes method. These methods do not generate summaries according to the user's interest. One of the methods of summarizing videos works by detecting shots by measuring the transition between successive frames.

According to the studies in (ZHANG et al., 2016) and (MAHASSENI; LAM; TODOROVIC, 2017), video summarization is treated as a seq-to-seq problem which produces the keyshot/keyframe sequence. The understanding of heterogeneous interdependency between video frames plays an important role in generating accurate video summaries. LSTMs incorporated with encoder and decoder frameworks were used in both studies to model variable range dependency. Since LSTMs are known to solve the vanishing gradient problems posed by RNNs, both (ZHANG et al., 2016) and (MAHASSENI; LAM; TODOROVIC, 2017) achieved state-of-the-art results. But this solution comes with one shortcoming that all the input frames are given the same importance regardless of the output shots to be predicted. This is because all the necessary information is encoded in one single context vector. To aid this problem, both studies simply ignored the temporal structure of the video.

3. DATASET

Instead of benchmark datasets of video summarization; TVSUM and SUMME, Ego4D dataset is used for this project because query focused video summarization requires the natural language representations of videos in order to produce video summaries guided by the user query which is also given in natural language.
Ego4D is a large-scale, multi-modal dataset for egocentric (first-person) video understanding. It consists of over 3,000 hours of video recordings from 700 unique individuals, capturing daily activities like cooking, cleaning, and socializing. The dataset includes:

1. 3,000+ hours of egocentric video.
2. 40,000+ annotated episodes (short video clips).
3. 12,000+ hours of audio narration.
4. 2.5 million+ annotated objects, actions, and events.

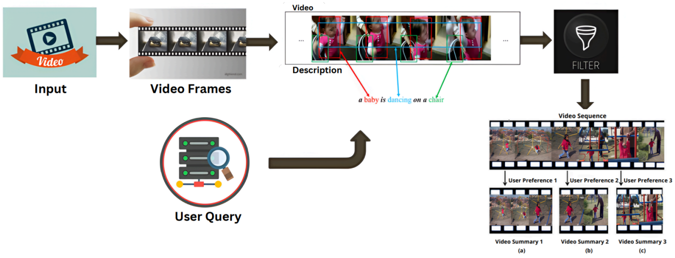
The Ego4D dataset has been limited to the use case of cleaning and laundry videos for this project. Total 44 videos of varying length (3 to 30 minutes) were downloaded and used for training the captioning model. The data contains the description of frames, frame number and time stamp along with the video.

4. METHODOLOGY

The process of video summarization, as shown in Figure 1, entails two primary tasks: image captioning and retrieving frames similar to a user query. For this purpose, the following steps were performed:

1. A given video (30 FPS) is broken down into its constituent frames.
2. All those frames were selected for training whose descriptions were given in the dataset.
3. These frames are saved as numpy arrays and PIL Images.
4. A hugging face dataset is created which contains frames and their respective descriptions.
5. A combination of two large pretrained models (ViT and GPT-2) is finetuned over this custom dataset.
6. For each video, a description is generated every 15th frame (2 frames' descriptions generated per second).

Figure 1. High Level Framework



7. The set of these generated descriptions along with their respective frame numbers are made into documents of LlamaIndex.
8. These nodes are stored into VectorStoreIndex which acts as a retriever.
9. The retriever is then queried with the user query, and it outputs top 60 frames similar to the user query.
10. The overlapping frames are eliminated to avoid repetition of frames in a summary.
11. For every similar frame preceding and succeeding 4-second segments are selected alongside them to maintain coherence within the summary.
12. The generated summaries are evaluated against human generated ground truth. The evaluation metrics used are recall and F1-score.

Mathematical Formulation

$P = \{V_0, V_1, \ldots, V_n\}$
$Where,\ n \in 1, 2, ..., N$ & P = Video from a set of videos.

A video can be represented as a sequence $V_i = \{f_0, f_1, \ldots, f_n\}$
$Where,\ t \in 1, 2, ..., N$ & n = number of frames.

A set of queries inputted by the user can be represented as $Q = \{q_0, q_1, \ldots, q_m\}$
$Where,\quad m \in 1,2,..., M$

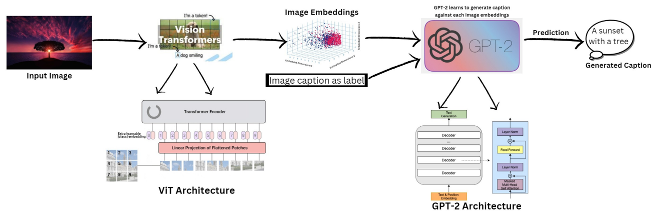
given that,
$$m << n$$
and
$$|Q| = m$$

Model (P, q) = S (V_i)

Image Captioning By finetuning ViT-GPT2

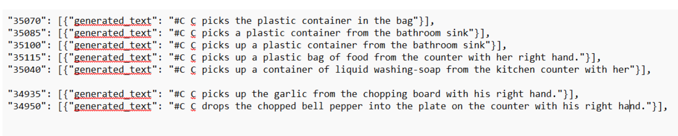
The decision to incorporate pretrained models like ViT and GPT-2 into our video summarization task was driven by their widespread acceptance and proven effectiveness. ViT was employed as the encoder to comprehend the visual content within the video and generate image representations, while GPT-2 served as the decoder to convert this information into natural language, producing frame-level descriptions by decoding the embeddings generated by ViT. Figure 2 summarizes this end-to-end workflow.

Figure 2. Image Captioning Model



Initially, the video was segmented into frames, and then the visual features of each frame were extracted using ViT's AutoFeatureExtractor, resulting in image embeddings, i.e., pixel values, for each frame. Simultaneously, GPT-2's AutoTokenizer was used to tokenize the descriptions of each frame. These image representations, along with their corresponding caption embeddings as labels, were fed into the GPT-2 model. These caption embeddings guided the learning process of GPT-2 to generate relevant frame descriptions. Consequently, descriptions of video frames were generated and saved with the respective frame number, as depicted in Figure 3, to retrieve frames pertinent to the user query by leveraging the similarity between the generated frame-level descriptions and the user query.

Figure 3. Generated Captions from the aforementioned Model



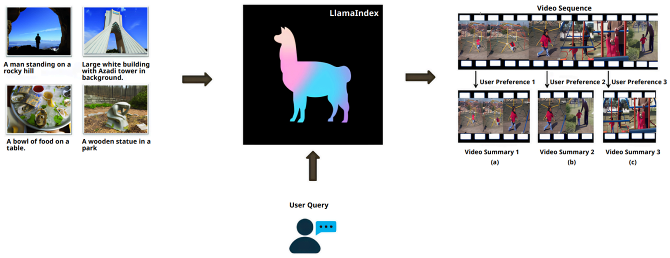
```
"35070": [{"generated_text": "#C C picks the plastic container in the bag"}],
"35085": [{"generated_text": "#C C picks a plastic container from the bathroom sink"}],
"35100": [{"generated_text": "#C C picks up a plastic container from the bathroom sink"}],
"35115": [{"generated_text": "#C C picks up a plastic bag of food from the counter with her right hand."}],
"35040": [{"generated_text": "#C C picks up a container of liquid washing-soap from the kitchen counter with her"}],

"34935": [{"generated_text": "#C C picks up the garlic from the chopping board with his right hand."}],
"34950": [{"generated_text": "#C C drops the chopped bell pepper into the plate on the counter with his right hand."}],
```

■ Similar Frame Retrieval Using LlamaIndex

Descriptions of video frames are converted into documents by extracting frame numbers and corresponding generated text. As illustrated in Figure 4 each text is then encapsulated within a document along with its associated metadata, which includes the frame number. Subsequently, a LangchainNodeParser is employed to parse nodes from the documents, utilizing a RecursiveCharacterTextSplitter.

Figure 4. Frame Retrieval



During this parsing process, the embeddings model from hugging face is utilized to generate embeddings for documents list, thereby constructing structured nodes. These nodes are then used to construct a VectorStoreIndex, which organizes the embeddings in a manner conducive to efficient similarity search operations. The VectorStoreIndex is converted into a retriever, as depicted in Figure 4, with a specified parameter (20, in our case) for the number of similar items to be retrieved with respect to the given user query.

5. EVALUATION RESULTS

5.1. Rouge

The Recall-Oriented Understudy for Gisting Evaluation (ROUGE) score measures how similar a candidate document is to a collection of reference documents. In this project, Rouge is used during finetuning of the ViT-GPT-2 model. It evaluates the model's performance by calculating the similarity between the generated caption and input frame's caption.

Scores obtained for 2 epochs during training are as follows:

5.2. Bleu Score

The BiLingual Evaluation Understudy (BLEU) is a metric used to evaluate machine translated text. It ranges from zero to one and indicates how similar

Table 1. Training Results

Rouge1	Rouge2	Rougel	Rougelsum
48.994500	27.722300	48.317200	48.403600
54.334000	34.359200	53.537800	53.532900

the machine translated text is to a set of reference translations. If machine translation is shorter than the reference translation, a Brevity penalty is applied to the precision score.

On 1040 testing images bleu score was calculated 0.512 value was achieved.

5.3. Recall & F1 Score

For evaluating the machine generated video summaries, commonly used metrics are recall and F1-score. True positive, false positive and false negatives, in this case, would be:

True Positives = Frames in both ground truth and machine summary.
False Positives = Frames in machine summary but not in ground truth.
False Negatives = Frames in ground truth but not in machine summary.

Video Summaries were generated for 5 videos and evaluated against human generated ground truth summaries. The obtained F1-score was 47.73, an increase of 4% from the baseline (see Figure 5) is observed on this dataset.

Following is the evaluation table of 5 videos:

Table 2. Testing Results

S.No.	F1 Score	Recall
1.	41.74	54.64
2.	71.34	70.50
3.	19.79	35.02
4.	60.06	75.17
5.	45.74	51.48
Average	**47.73**	**57.36**

6. COMPARISON

Paper	Model	Dataset	F1-Score	Recall
Query-Focused Extractive Video Summarization	Sequential & Hierarchical Determinantal Point Process (SH-DPP)	UT Egocentric & TV Episodes	~ 21 - 37	~ 36 - 45
Query-Focused Video Summarization: Dataset, Evaluation, and A Memory Network Based Approach	Memory network-based model, parameterized by a Sequential Determinantal Point Process (SeqDPP)	UT Egocentric	44.19	60.25
UniMD: Towards Unifying Moment Retrieval and Temporal Action Detection	UniMD framework consists of a shared encoder for feature extraction, task-specific heads for moment retrieval (MR) and temporal action detection (TAD), and multi-task learning to optimize both MR and TAD losses simultaneously	Ego4D	-	44.8
ViT-GPT2 Model (Our Model)	ViT & GPT-2 for image captioning. WHereas, LlamaIndex for similar frame retrieval.	Ego4D	**47.73**	**57.36**

Figure 5. Performance Comparison

7. CONCLUSION

This research presents a hybrid approach to user-specific video summarization, combining GPT-2, Vision Transformers (ViT), and LlamaIndex. By leveraging the power of pretrained models, our method efficiently generates personalized video summaries based on user queries, reducing the need for extensive computational resources. The approach was tested on the Ego4D dataset, demonstrating promising results with an F1 score of 47.7 and recall of 57.3. These findings highlight the effectiveness of combining advanced language and vision models for tailored video summarization. Future work includes enhancing the model's ability to handle video captions more effectively and expanding its application to broader datasets.

7.1. Future Work

1. The image captioning model should be turned into a video captioning model. This would also require the model to maintain the context of some previous frames to generate a caption of the current frame accordingly.
2. More can be achieved if the model learns to consider the textual information present within a video frame. This type of video summarization would be very beneficial in educational sectors, where there are video frames loaded with textual information.
3. The model is trained only on cleaning and laundry videos. To further broaden this umbrella, a dataset (videos with their frame level descriptions) must be prepared for that domain.
4. Object detection using YOLO can identify the objects present in the video frames. This will help in retrieving frames having the objects similar to those specified by the user in his query.

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